

De novo protein design for chemical recognition and enzymatic catalysis

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Recent breakthroughs in AI-based protein structural biology have transformed protein structure prediction and protein design. However, design challenges that require atomic-level accuracy and high-resolution modeling, such as small-molecule binding protein design and enzyme design, remain unsolved. To address these challenges, we used RFdiffusion All-Atom, a generative AI model fine-tuned from RoseTTAFold All-Atom. RoseTTAFold all-atom was developed to predict the structures of diverse biomolecular complexes. We also developed LigandMPNN to generate sequences conditioned on protein-bound small-molecule coordinates. By combining these methods, we designed de novo binders for eight distinct small-molecule targets, including metabolites and small-molecule drugs. The binders were readily expressed in *E. coli* and exhibited binding affinities ranging from the nanomolar to low-micromolar range. Protein-ligand co-crystal structures validated the designs, showing close agreement between the experimental structures and the computational models. Building on this success in designing interactions between proteins and chemical groups, we are now extending our methods toward allosteric sensor design, de novo enzyme design, protein-RNA interface design, and binder design for recognizing chemical modifications such as glycosylation, phosphorylation, and methylation.